

Hide menu

3.1: Introduction to the course

Reading: Frequently Asked Questions

10 min

Reading: Course Resources

10 min

Reading: How to Use Discussion Forums

10 min

Discussion Prompt: Introduce yourself

10 min

Reading: Earn a Course Certificate

10 min

Reading: Are you interested in earning an online MSEE degree?

10 min

3.1.1: Welcome to the course!

3.1.2: Introducing nonlinear variations to Kalman filters

3.1.3: Deriving the three extended-Kalman-filter prediction steps

3.1.4: Deriving the three extended-Kalman-filter correction steps

3.1.5: Introducing a nontrivial EKF example, finding derivatives

3.1.6: Introducing Octave code to initialize and control EKF for state estimation

3.1.7: Introducing Octave code to update EKF for state estimation

3.1.8: Summary of "The extended Kalman filter" module plus next steps

Course Resources

Suggested Reading

While there is no required textbook for this course, the textbook I use in my graduate course at UCCS is:

- [Optimal State Estimation: Kalman, H Infinity, and Nonlinear Approaches](#) by Dan Simon

I believe that this textbook provides a good permanent reference to the material covered in this course, and will enable further study (it will be used by the other courses in the specialization as well).

Octave

You will be using the Octave programming language to complete quiz requirements in this course and in other courses in the specialization. Much of the Octave code that you will need will be provided to you, but you will be required to modify this code and to write some of your own. The parts of Octave that we will use are nearly 100% compatible with the syntax of MATLAB – if you already know either Octave or MATLAB, you are well prepared.

If you do not know either MATLAB or Octave, there are many tutorials on the Internet available to help with the basics you will need. For example, you might confer this [Octave tutorial](#).

Learner Support

The nature of online learning and Kalman filters are both things that involve technical components, so it is to be expected that some of you may experience issues. With a bit of effort, you will find the answers to your questions right under your nose (hint – check the forums). If you find yourself having specific problems, please refer to these resources:

Course Material

The course forums are a great way to connect with your peers and get answers to any questions you may have throughout the course.

Coursera

If you have issues on the Coursera platform, please check out [Learner Support](#). You may also contact Coursera directly [here](#).

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